MEMORY CIRCUITS AND SYSTEM DESIGN

COURSE PROJECT:

DRAM Engineering Lab:



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Problem Statement:

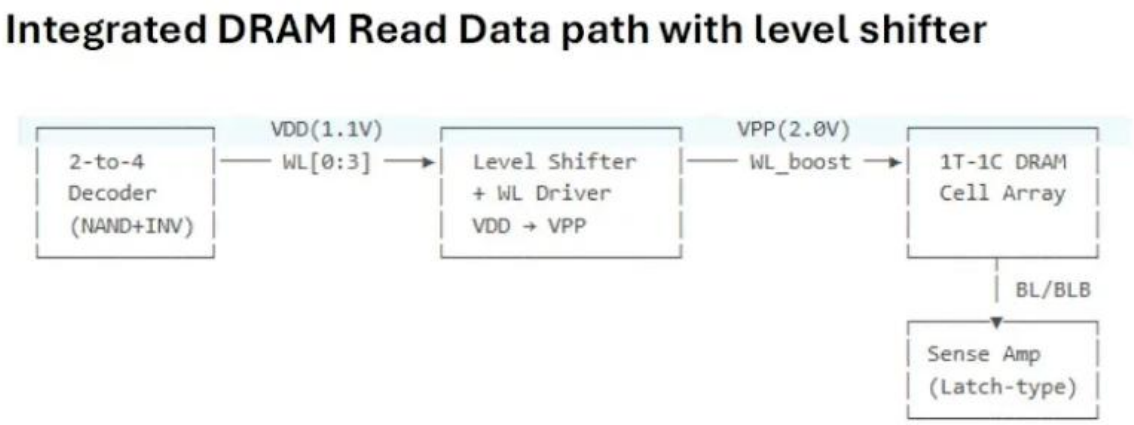
Design and simulate the core read path of a DRAM array consisting of three key building blocks:

1. A 1T-1C DRAM Cell demonstrating charge sharing

2. A Sense Amplifier to detect and amplify the small differential bit-line voltage

3. A 2-to-4 Row Decoder to select the word-line (Do you think we need a level shifter? If so, what is the reason)

Integrate all the blocks to understand how these three blocks interact during a DRAM read operation from cell access through signal sensing to full-swing output. [Building of the level shifter can be optional]



Objective:

The objective of this lab exercise is to design, integrate, and simulate the critical components of a Dynamic Random Access Memory (DRAM) read path. The read path forms the most critical timing chain in memory design, spanning from the activation of a Row Decoder to the development of a full-swing logic output by a Sense Amplifier. Understanding this interaction involves analysing the sequential processes of row selection, charge sharing between the storage cell and bit-line, and differential signal amplification.

1T-1C DRAM Cell and the Charge Sharing Principle

The fundamental storage unit in a DRAM array is the 1T-1C cell, comprising one Access Transistor (NMOS) and one Storage Capacitor Data is stored as charge on a "High" voltage represents a Logic ‘1’, and "Low" voltage (Ground) represents a Logic ‘0’.

The core mechanism of a read operation is **Charge Sharing**. The process occurs in specific phases:

1. **Pre-charge Phase:** Before reading, the Bit-line (BL) capacitance C_{BL} is pre-charged to a reference voltage, typically halfway between supply and ground ($V_{PRE} = V_{DD}/2$). The complementary bit-line (BL^{B}), often connected to a

reference dummy cell, is also held at V_{PRE}. The Word-line (WL) is held Low (Ground), keeping the access transistor OFF.

2. **Access Phase:** The Row Decoder activates the specified Word-line (raises it High). This turns ON the access transistor, connecting the storage capacitor C_{cell} to the pre-charged Bit-line (C_{BL}).

3. **Charge Sharing:** Charges redistribute between C_{cell} and C_{BL} until reaching equilibrium. This causes a very small voltage change on the bitline.

    - **Reading '1':** If C_{cell} stored V_{DD}, charge flows from the cell to the bit-line, causing V_{BL} to rise slightly above V_{DD} /2.
    - **Reading '0':** If C_{cell} stored Ground, charge flows from the bit-line to the cell, causing V_{BL} to drop slightly below V_{DD} /2.

The new bitline voltage (V_{BL_final}) is determined by the conservation of charge:

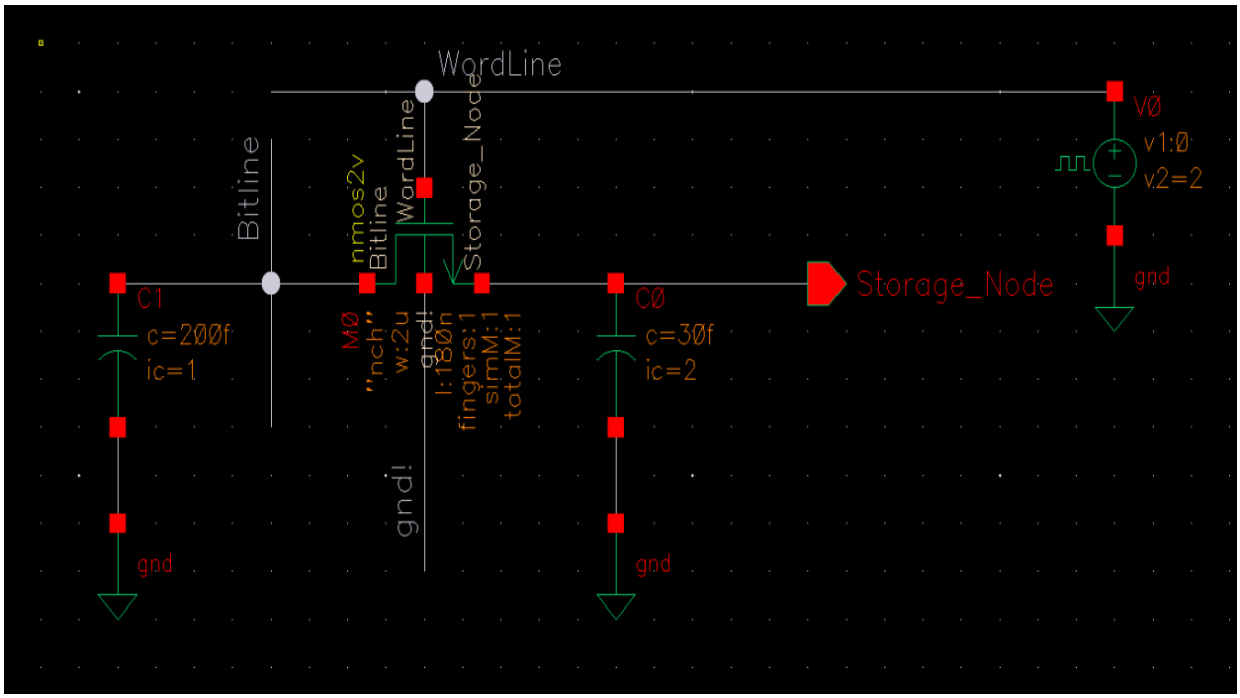
$$V_{BL_final} = \frac{C_{BL} \cdot V_{PRE} + C_{cell} \cdot V_{stored}}{C_{BL} + C_{cell}}$$

The developing differential voltage seen by the sense amplifier is approximately:

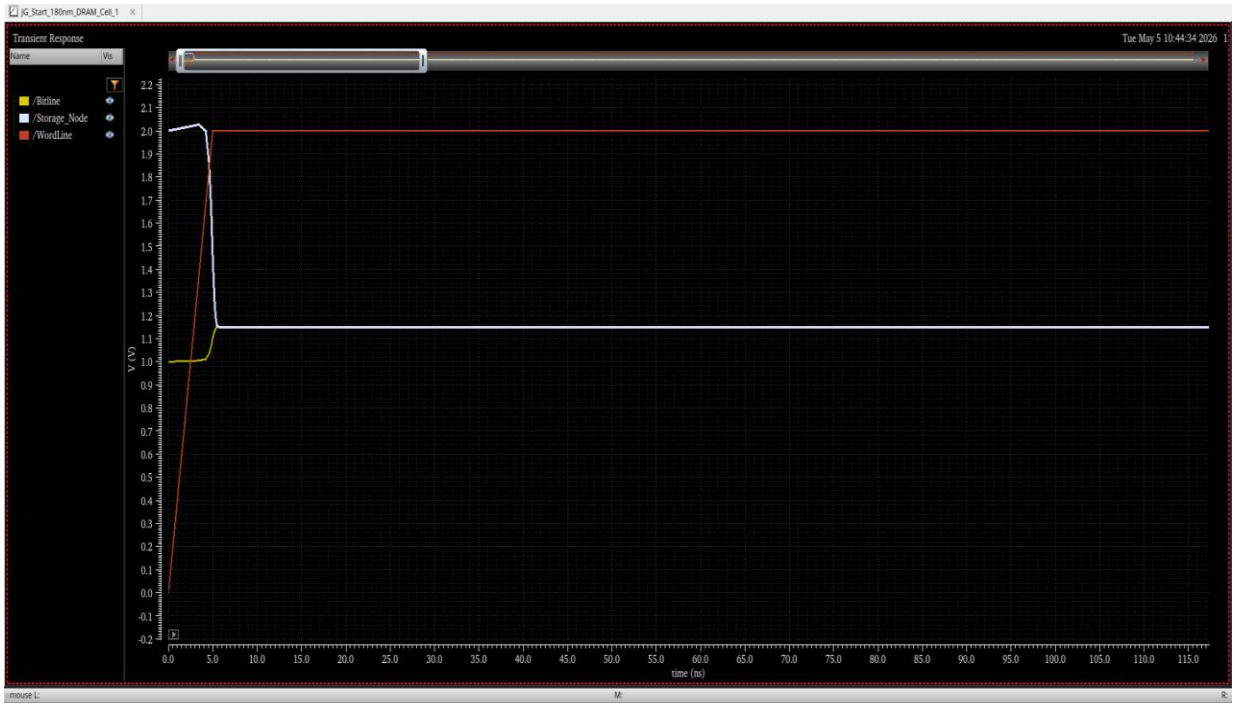
$$\Delta V = |V_{BL_final} - V_{PRE}| \approx \left(\frac{C_{cell}}{C_{BL} + C_{cell}} \right) \cdot |V_{stored} - V_{PRE}|$$

Because C_{BL} is typically much larger than C_{cell}, ΔV is very small (often in the tens to low hundreds of millivolts), demanding precise sensing.

Schematic Diagram:

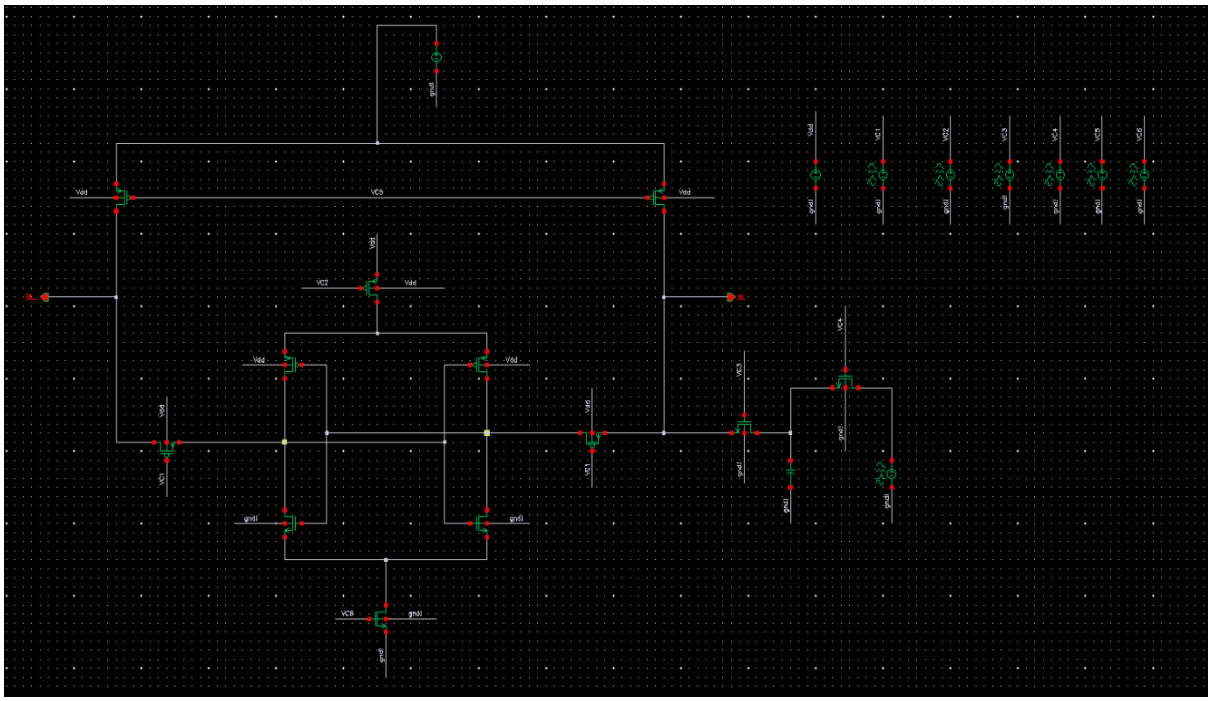


Simulation Result:



Sense Amplifier

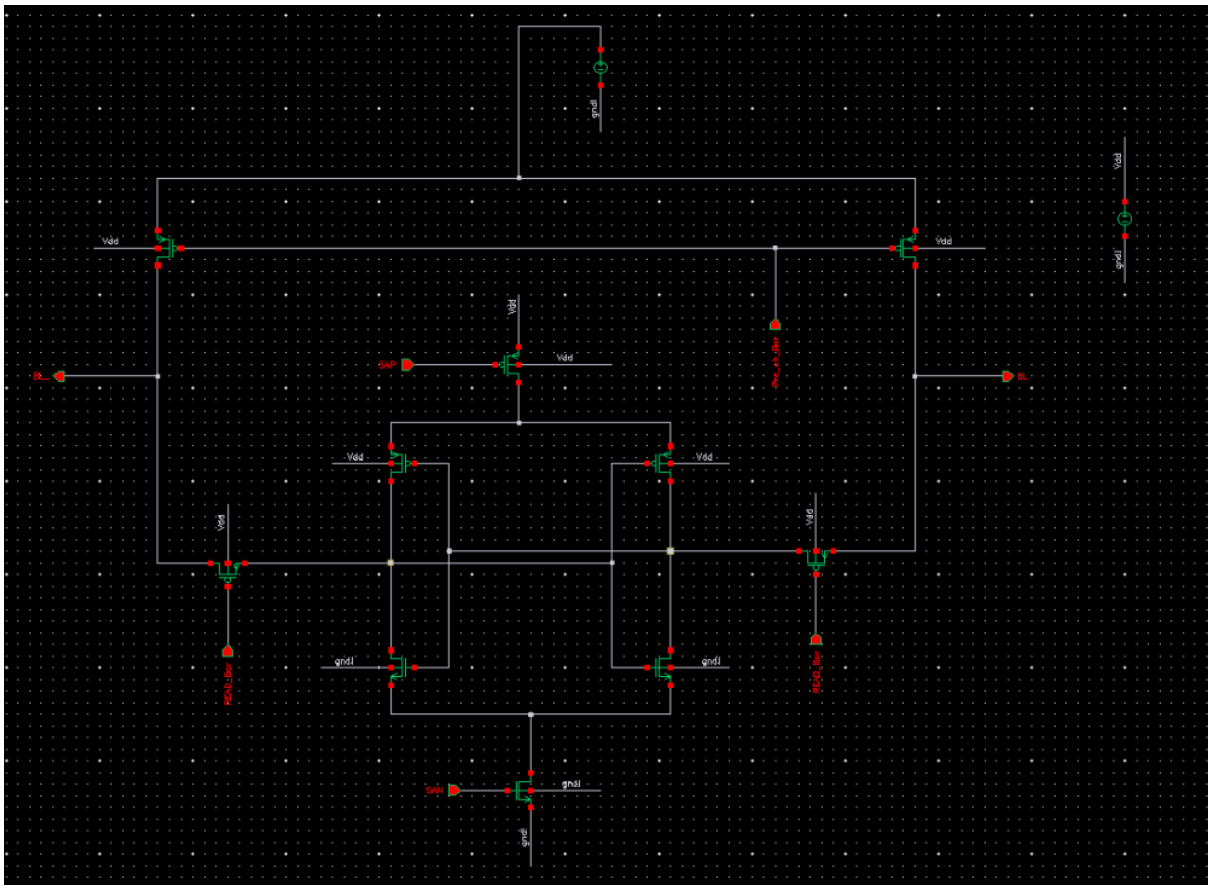
Testbench Schematic

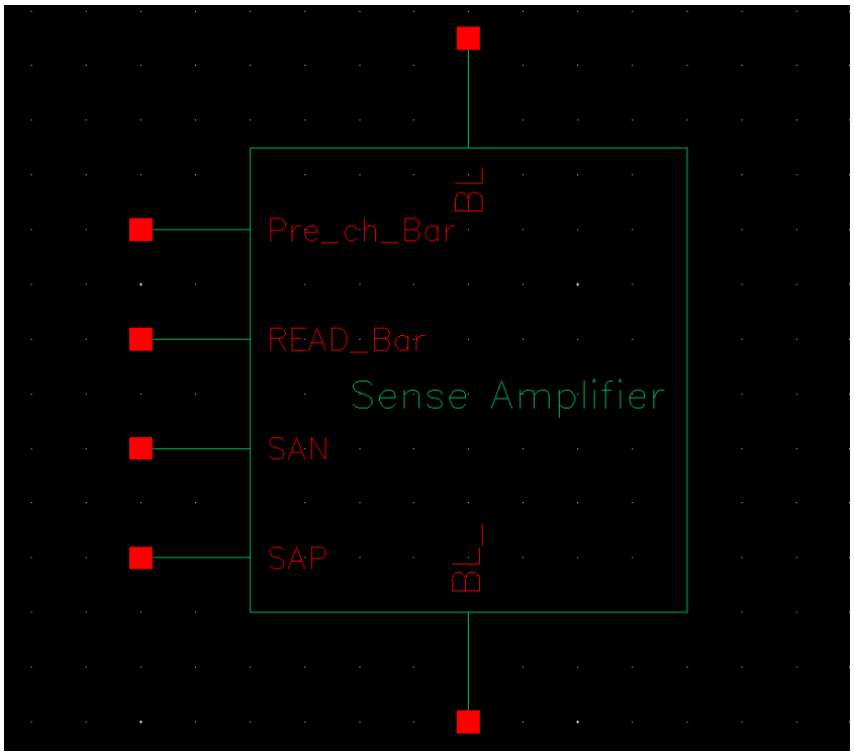


Simulation Output:



Schematic


Vdd
gnd!
SAP
BL
BL_
Pre_ch_Bar
READ_Bar


BL
Pre_ch_Bar
READ_Bar
Sense Amplifier
SAN
SAP
BL_

2-to-4 Row Decoder and Level-Shifter

A 2-to-4 row decoder is used to select one of the four word-lines in the DRAM array based on two input address signals (in_a0 and in_a1). The decoder generates four mutually exclusive outputs.

Only one word-line is activated at a time, enabling access to the corresponding DRAM cell connected to that row.

Need for Level Shifter

A level shifter is required between the decoder and the word-line driver.

The primary reason for using a level shifter is to overcome the threshold voltage limitation of the NMOS access transistor in the DRAM cell. If the word-line voltage is limited to VDD, the access transistor cannot pass a full logic high level to the storage capacitor due to the threshold voltage drop. As a result, the capacitor cannot be fully charged, leading to reduced stored charge and degraded read performance.

To ensure proper operation, the word-line voltage is boosted above VDD (typically close to VDD + V_t). This allows the access transistor to turn on strongly and enables full charge transfer between the bit-line and the storage capacitor, improving the sensing margin and reliability of the DRAM read operation.

Implementation

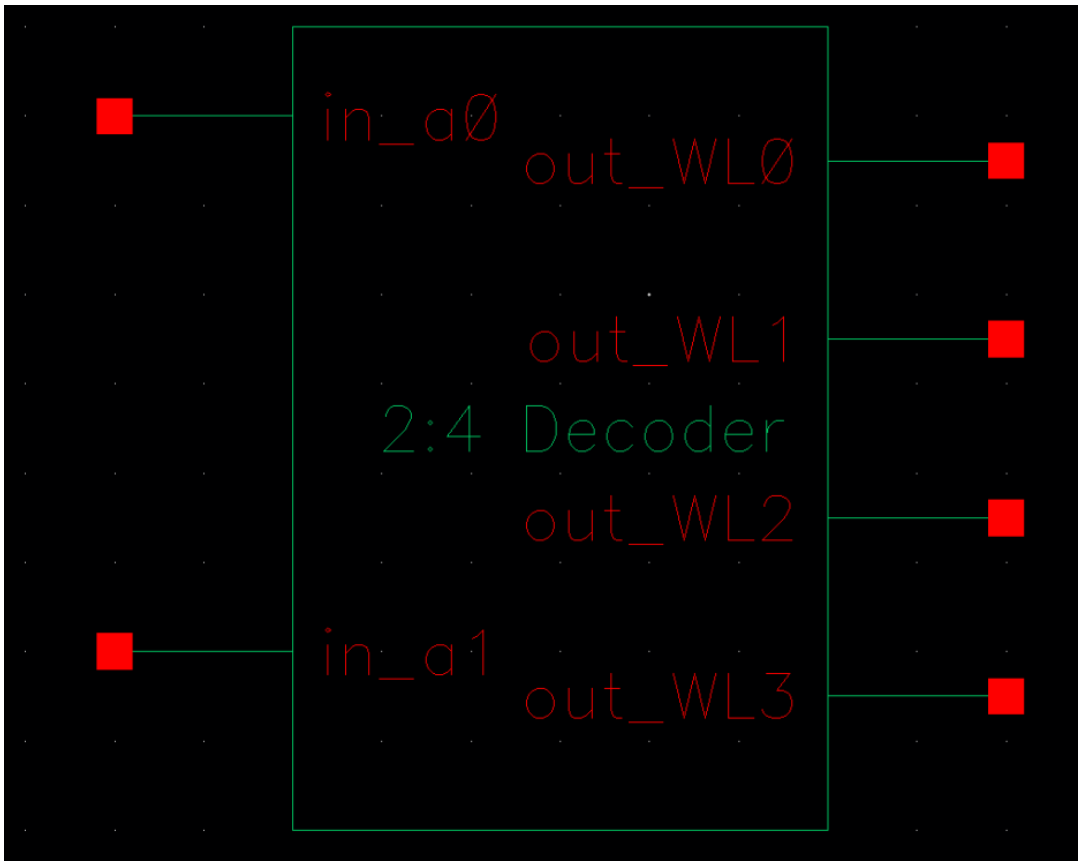
A cross-coupled level shifter is used to convert the low-voltage decoder output to a higher voltage suitable for driving the word-line. The circuit consists of cross-coupled PMOS transistors connected to the higher supply (2 V) and NMOS transistors driven by the input signal.

In the implemented design, the level shifter successfully converts the input signal to the higher voltage domain. The output is observed to swing up to VDD = 2 V when the input is asserted, confirming correct level shifting operation. This boosted output is then suitable for driving the DRAM word-line.

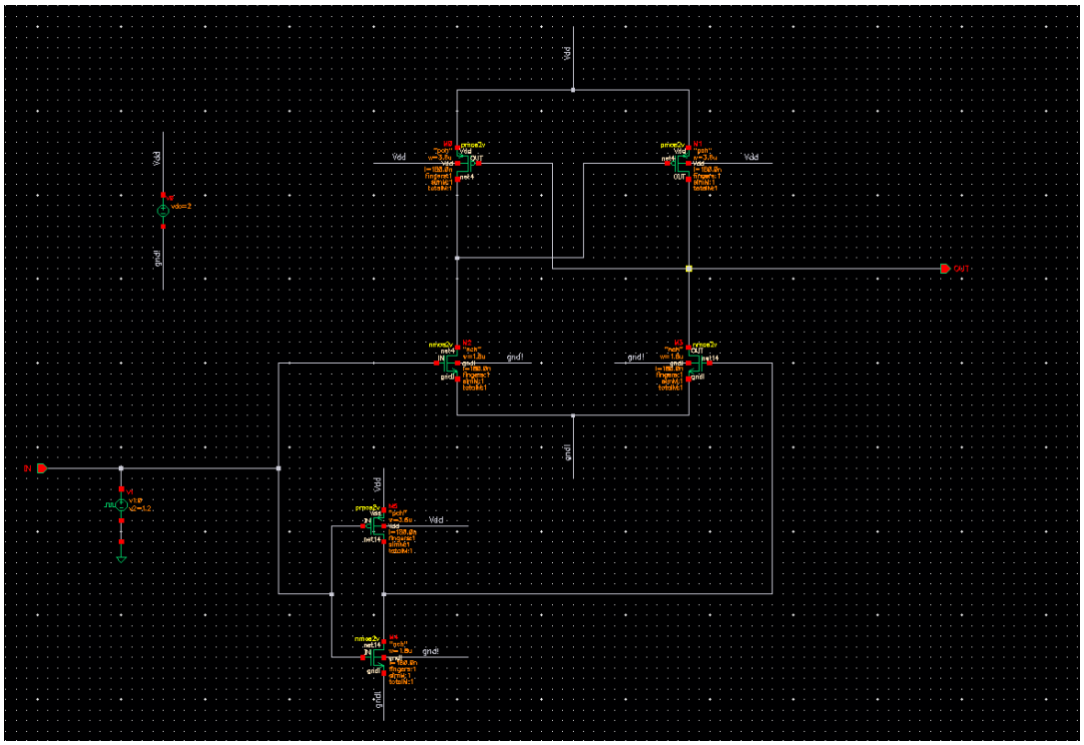
Conclusion

A level shifter is essential in DRAM design to bridge the gap between low-voltage logic circuits and high-voltage word-line requirements. It ensures full charge transfer, improves sensing accuracy, and enables reliable read and write operations in the memory cell.

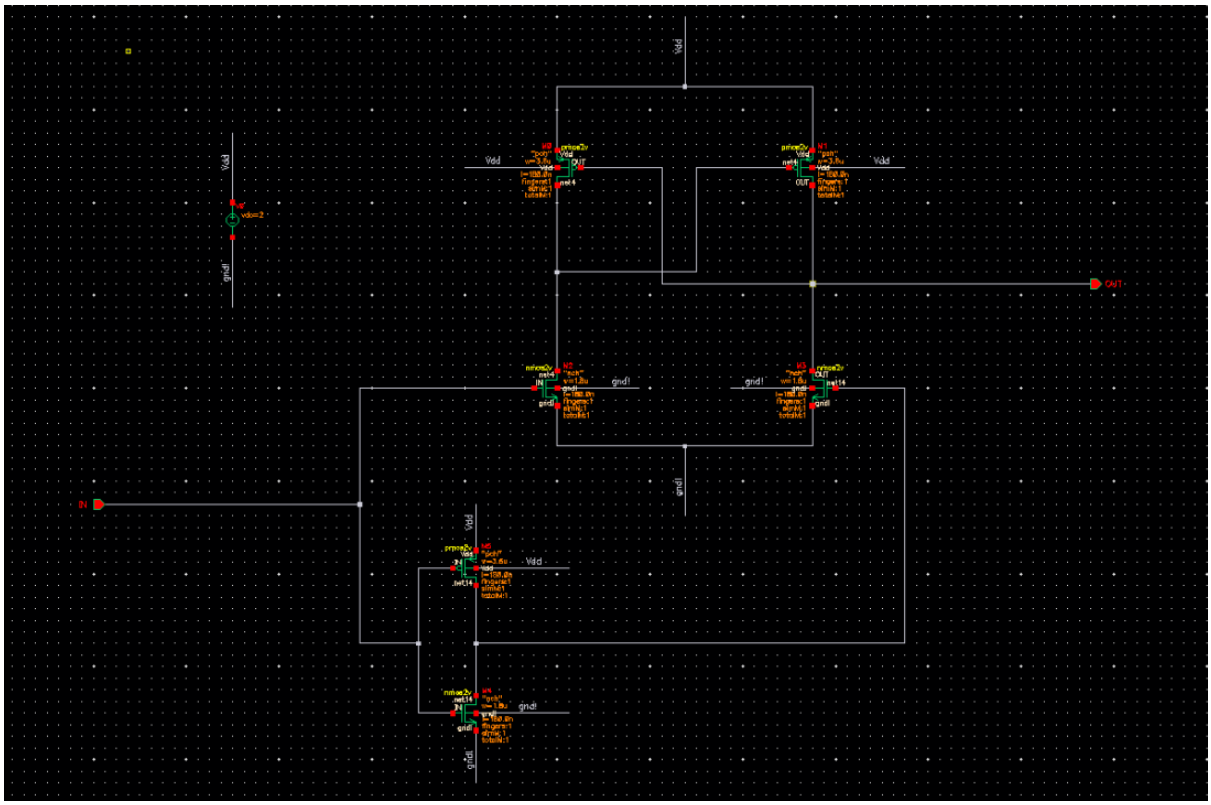
Schematic Diagram:

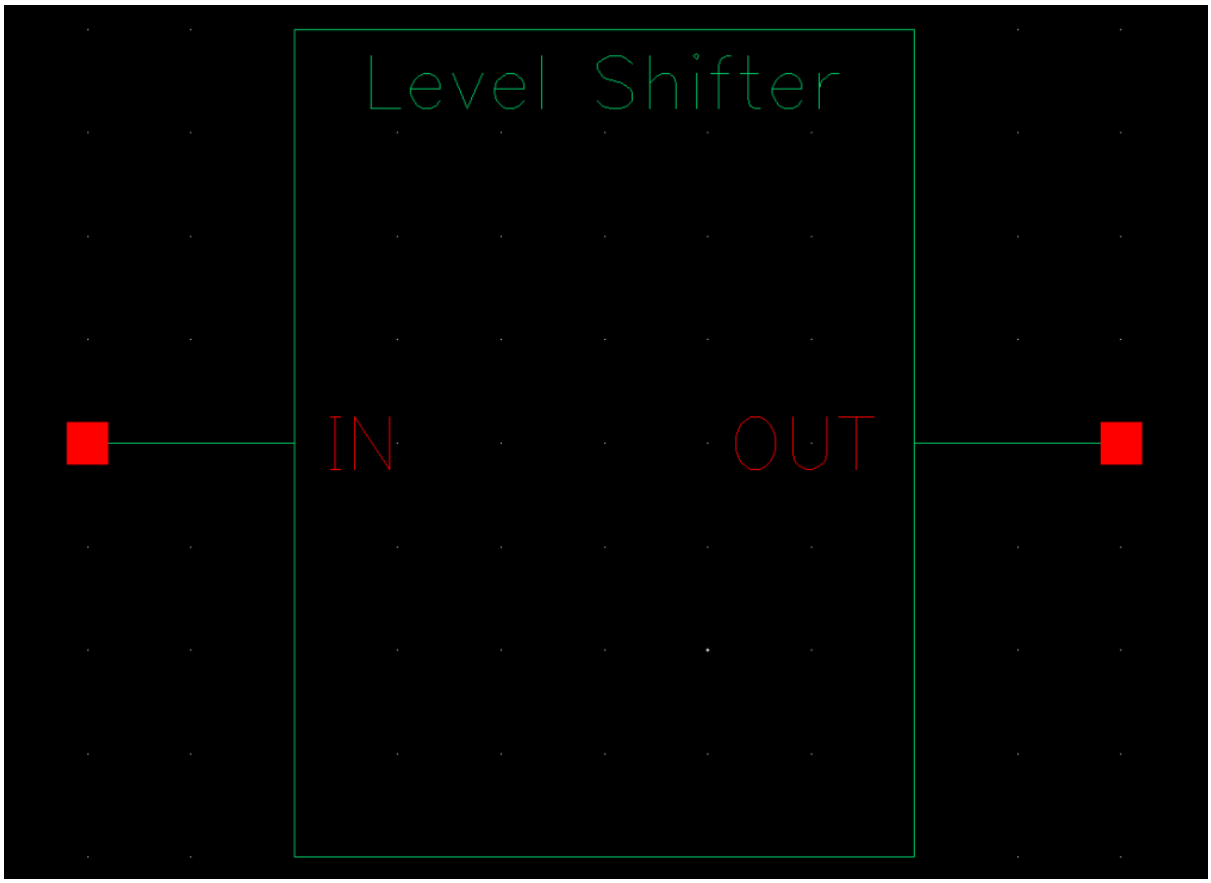


Level Shifter Testbench

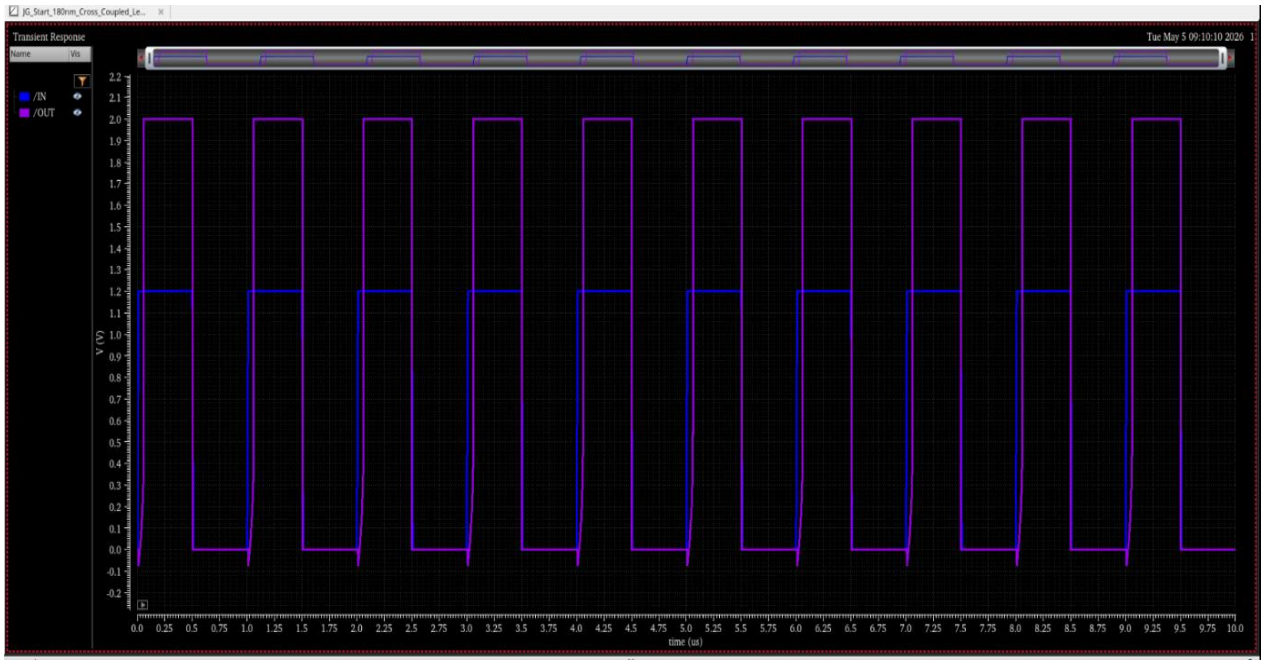


Level Shifter

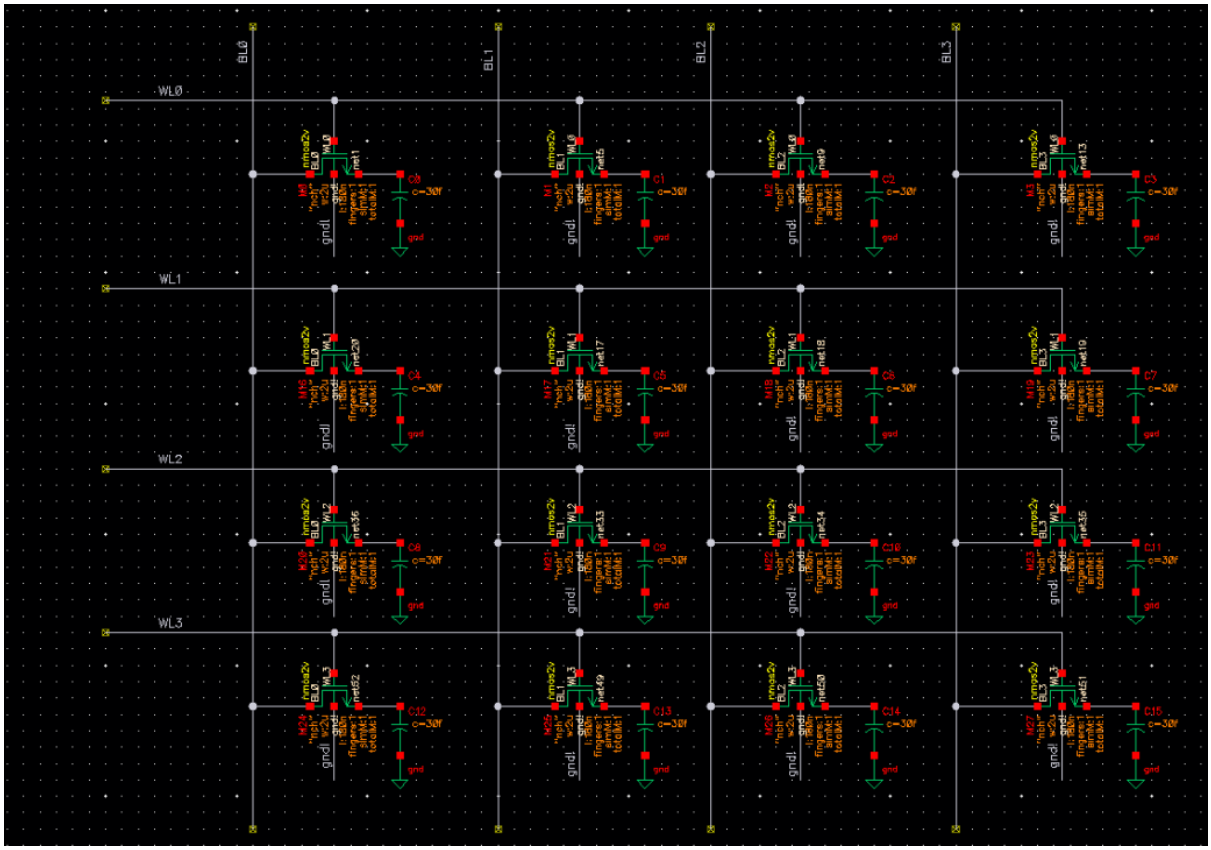




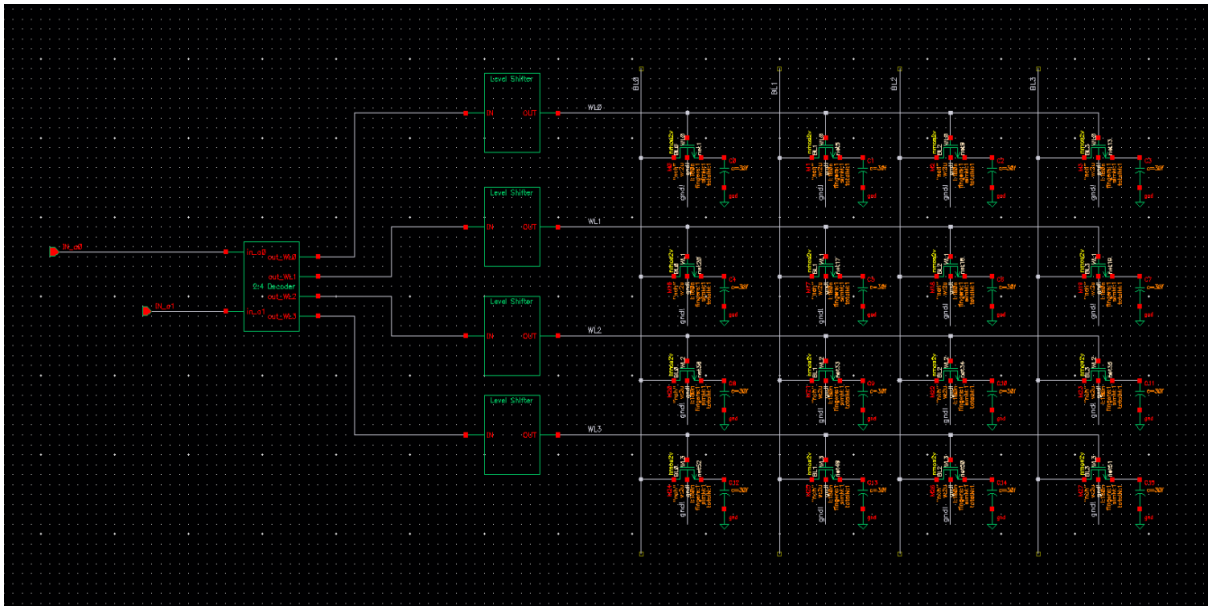
Simulation Result:



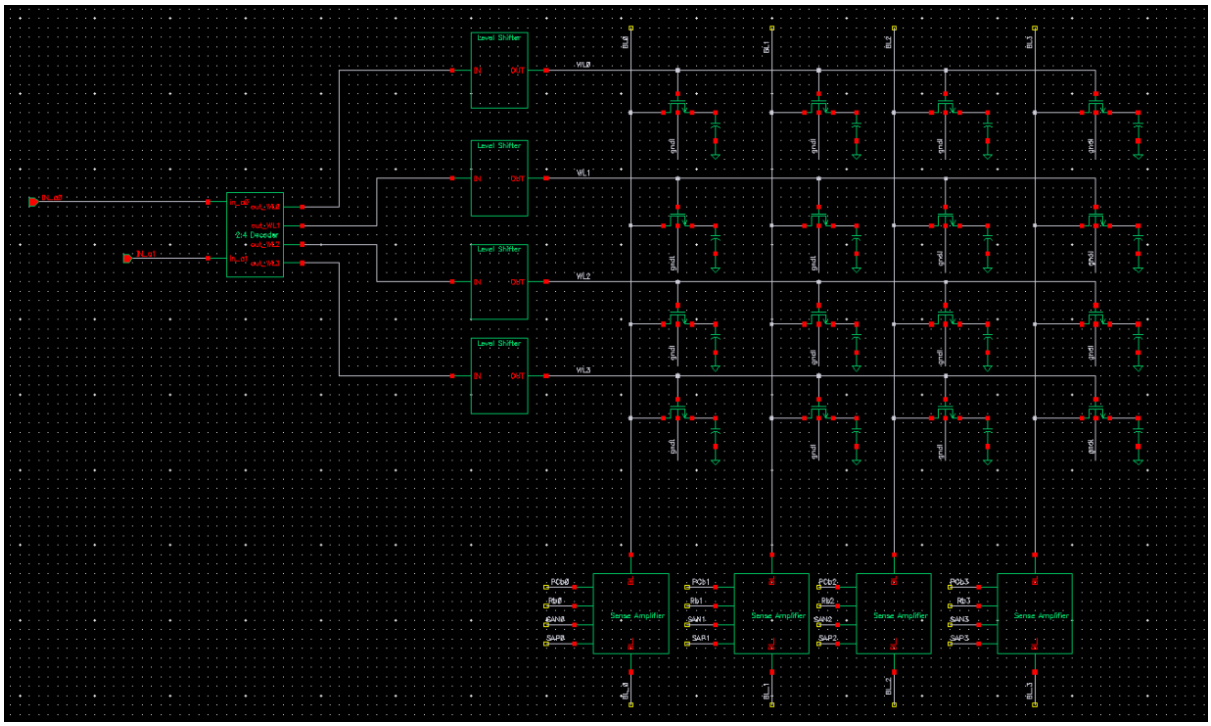
DRAM Array – 4 × 4



DRAM Array with Decoder and Level Shifter



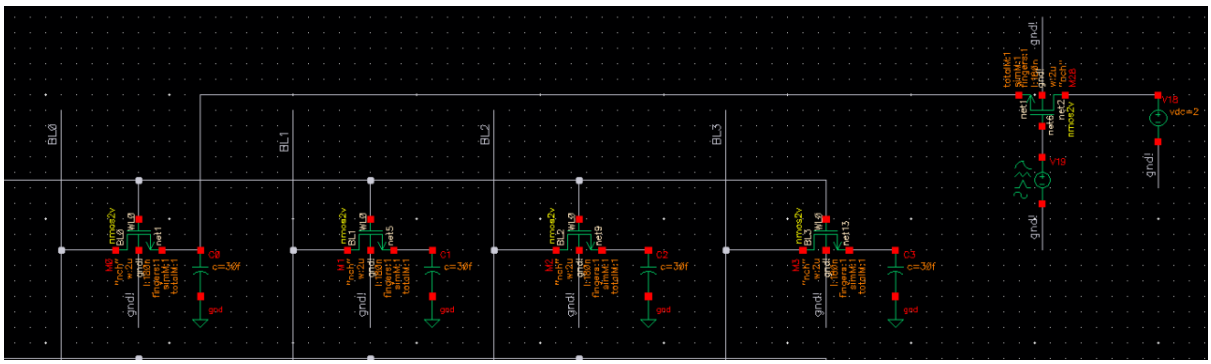
Completely Integrated DRAM Array



Integrated DRAM Array – Testbench to Read DRAM Array (0,0)

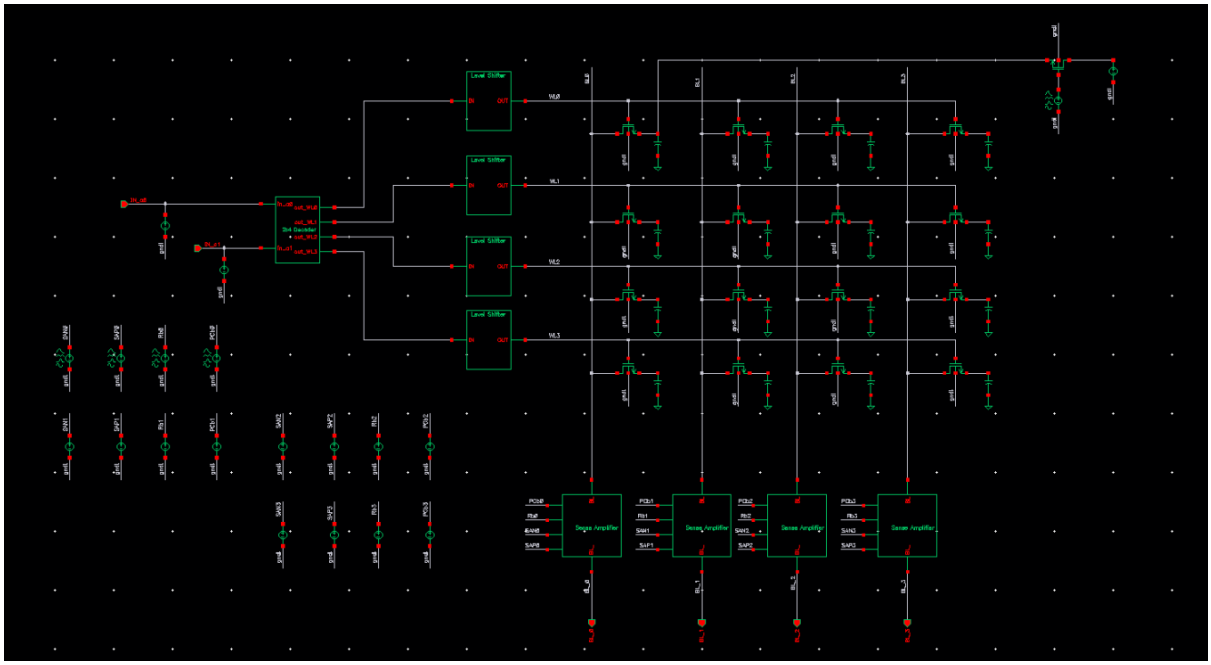
(BL0 and WL0)

The DRAM cell capacitor at location (0,0,) is initially charged to logic 1 (Vdd = 2V) using a circuit shown below:

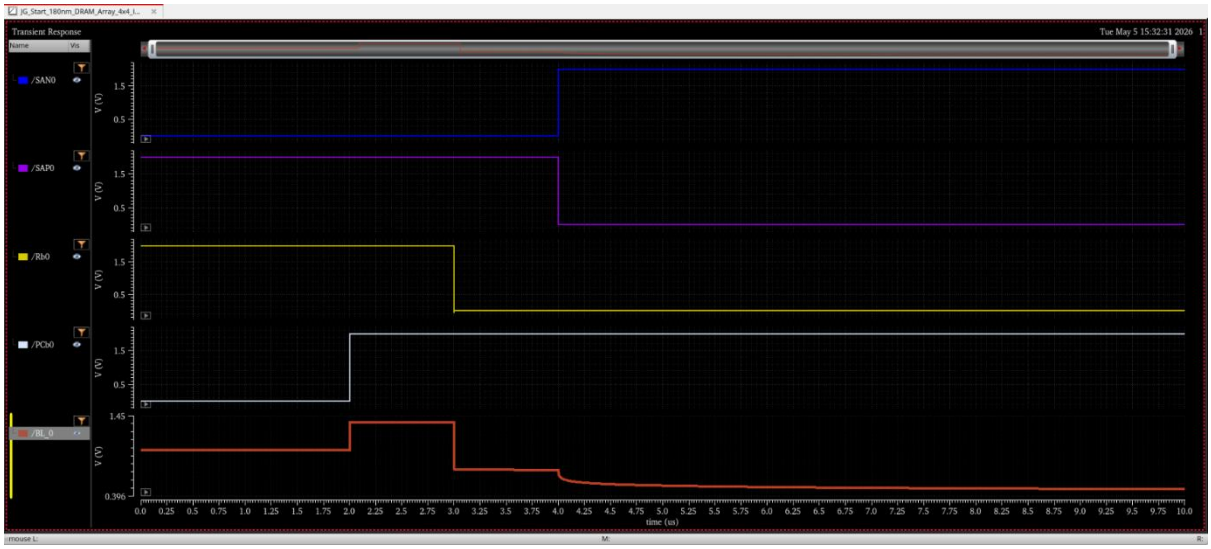


and then the cap node is left floating before starting the DRAM Read Operation

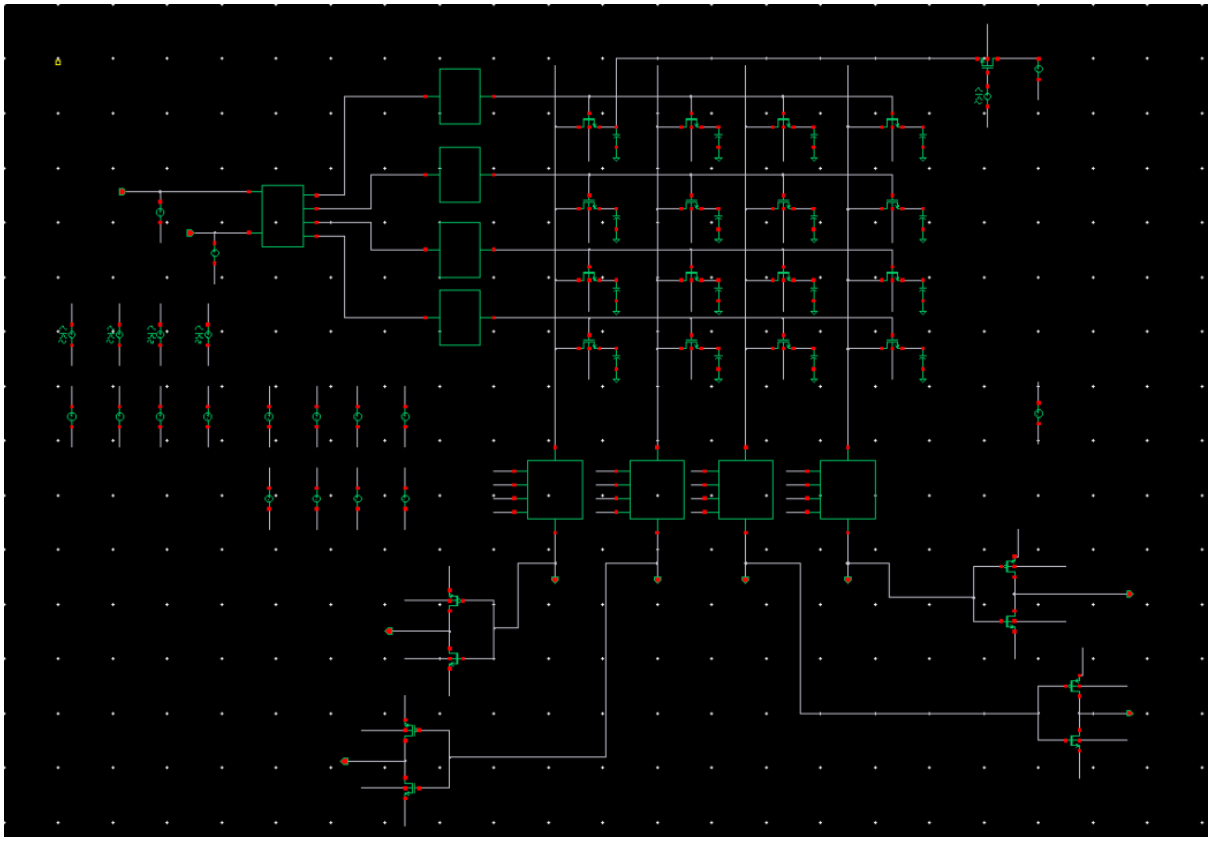
Testbench Schematic:



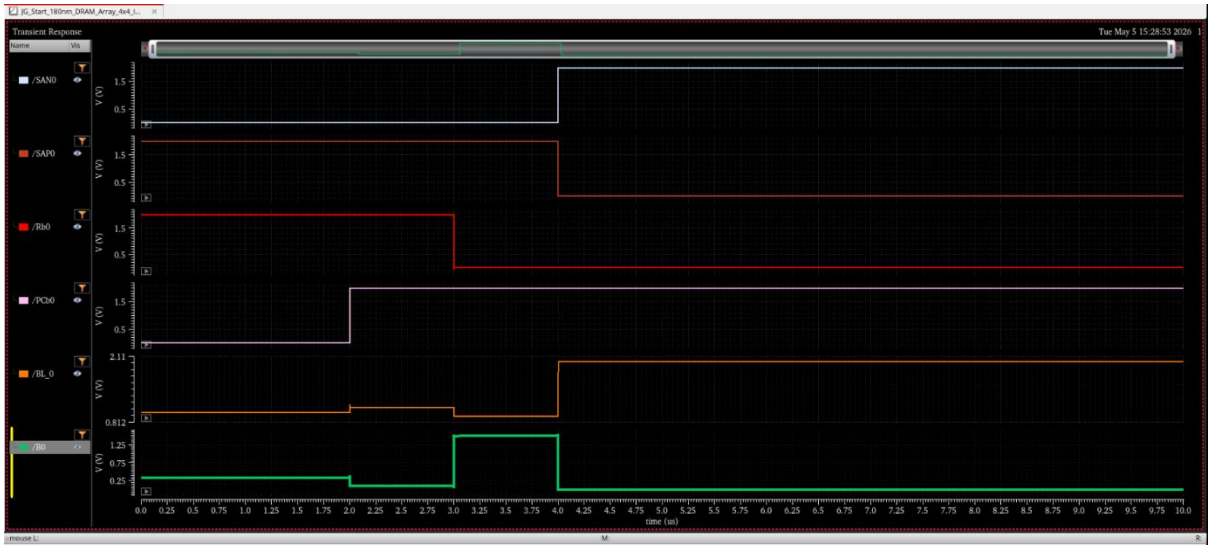
Simulation Output:



Testbench Schematic (Inverters are used to get the Bit back):



Simulation Output:



Conclusion

In this work, a DRAM read path was successfully designed and simulated using 180 nm technology. The system includes a 1T–1C DRAM cell, a 2-to-4 row decoder, a cross-coupled level shifter for word-line boosting, and a latch-type sense amplifier. The simulations demonstrate correct charge sharing between the storage capacitor and bit-line, followed by amplification of the small voltage difference using the sense amplifier.

The level shifter effectively converts the low-voltage decoder output to a higher voltage required for proper activation of the word-line, ensuring reliable access to the cell. The overall design validates the fundamental DRAM read operation, including word-line selection, charge redistribution, and signal sensing.

This implementation highlights the importance of voltage boosting, timing control, and differential sensing in achieving accurate and robust DRAM functionality.

